

Ontology(ies) of Data and Measurement

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- Availability of an enormous volume of data collected by heterogeneous sensors or resulting from complex analyses.
- Data circulate outside of the producers/collectors.
 - Interoperability: exchange data (information) and use the data (information) that has been exchanged.
 - Need to ensure the quality, reliability, validity, and trustworthiness of the analyses performed on data from different sources.
- Data integration.
 - *Unification* of data sources.
 - Build a single (consistent) body of evidence.
 - *Linkage* of data sources.
 - Link and compare datasets reliably and efficiently.

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- eScience.
 - Large-scale, distributed collaborative science in which data exchange is often directed to ensure the **reproducibility** of scientific analyses and experiments.
 - Scientists use different terminologies, methodologies, instruments, and conceptual frameworks.
- Social Sciences · Cultural Heritage/Digital Humanities · eGov · Medicine.
 - Quantitative and qualitative data.
 - Direct and indirect data (produced by computation).
 - Involvement of humans in the collection/production phase.

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- Data as *propositional knowledge*: data are knowledge claims, they express knowledge about a domain of interest via classifications of entities (several entities) under concepts (relations).
 - Model of the domain.
 - *Ontologies* formally characterize the semantics of the primitive notions adopted to model the domain (+ offer reusable conceptual tools).
 - TBox: $60KG(x) \rightarrow WEIGHTS(x)$; $60KG(x) \rightarrow \neg 70KG(x)$
 - In relational DBs, the *data model* is mainly made explicit through documentation NL texts and table labels + *integrity constraints*.
 - Knowledge claims.
 - ABox: $60KG(john)$; $HEAVIER(john, mary)$.
 - Rows in a relational DB.
 - Knowledge Base: TBox + ABox (+ Inference system).

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- A domain can be conceptualized in different ways giving origin to different (possibly inconsistent) models/ontologies of a domain.
- Data sharing/access by integrating/aligning/using different models/ontos.
 - *DB schema integration* and *data warehouses* in the '80s.
 - *Ontology matching*: mappings between ontologies enable the (partial) transfer of data from one ontology to another.
 - *Ontology based data access / integration*: a single ontology used for queries together with mappings to the data sources (e.g., DBs)
 - “[provides] access to the data layer, consisting of autonomous data sources (e.g., DBs), through the mediation of a conceptual domain view, given in terms of an *ontology*, and the use of a declarative *mapping* between the data layer and the ontology.” (Xiao et al. 2018)
 - “[is] tasked to answer the user query by incorporating the information from the various information sources, possibly employing the domain knowledge in the ontology to infer new information” (Schneider and Šimkus 2020)

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- ('Provenance') "With a rapid increase in data from sensors being published on the Web, there is an increasing interest in the re-use and combination of that data. However, raw observation results do not provide the context required to interpret them properly and to make sense of these data. Searching, reusing, integrating, and interpreting data requires more information about the studied feature of interest"
- ('Sensors') "[W]ith today's diversity of data and data providers, notions that restrict the view of sensors to being technical devices need to be broadened. One example would be social sensing techniques such as semantic signatures to study humans and the data traces they actively and passively emit from within a sensor-observation framework. Simulations and forecasts are other examples showcasing why 'sensors' that produce estimates of properties in the world are not necessarily physical entities."
(Janowicz et al. 2018)

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- “Provenance of a resource is a record that describes entities and processes involved in producing and delivering or otherwise influencing that resource. Provenance provides a critical foundation for assessing authenticity, enabling trust, and allowing reproducibility. Provenance assertions (...) can themselves become important records with their own provenance.”
(Provenance Incubator Group 2010)
 - Allow future users of data to reconstruct the conditions under which they have been collected/produced.
 - Improve the understanding, sharing, and reuse of data.
 - Improve the trustworthiness of data sources.
 - Help solving conflicts between data and data sources, but we can also
 - apply judgment aggregation strategies, *e.g.*, *majority* rules;
 - consider trust on data sources, *e.g.*, $S1 <_T S2$ would rule out $S2$.

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- Measurement processes and theories are fundamental to understanding how data was acquired, which sensors and procedures were used, etc.
- Measurement Theories.
 - Representational theory of measurement (Krantz et al. 1971).
 - Epistemic theory of measurement (Frigerio et al. 2010, Mari et al. 2023).

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■ Representational theory of measurement (mathematical approach):

- recoding empirical observations in terms of familiar mathematical structures to use computational methods to infer facts about observations;
- Measurement: mapping from an *empirical* relational structure $\langle D, R_1, \dots, R_n \rangle$ to a *numerical* one $\langle N, S_1, \dots, S_n \rangle$, such that the relations among numbers (N) *represent* relations among entities (D).
 - “Without a well-defined homomorphism between an empirical and a numerical relational structure, it is far from clear how to interpret such numbers. [and numerical scales]” (Krantz et al. 1971)
 - What empirical structures can be represented by given well-known mathematical structures (similarities among different empirical structures).
 - Relations (operators) on N can be mapped back to relations on D .
 - ‘Numerization’ can imply an ‘abstraction’ process: numbers correspond to equivalence classes of entities indistinguishable in terms of $R_1, \dots, R_n$.
 - *Example*: (one dimension) rulers + concatenation represented by sum.
- **But...** empirical relational structures $\langle D, R_1, \dots, R_n \rangle$ are presupposed as given.

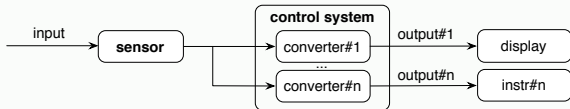
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■ Epistemic theory of measurement.

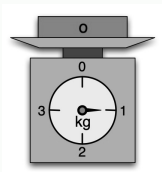
- “[A]n essential characterization of measurement is as an empirically structured model of the process, rather than some set of mathematical constraints on the inputs.” (Mari et al. 2023)
- Measurement
 - is an empirical process
 - designed on purpose
 - measurement processes implement *measurement procedures*
 - whose input is a property of an entity, and
 - measurement requires an **interaction** with an entity and that this interaction must be related to a property of that entity
 - that produces information in the form of **values** of that property.

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- Epistemic theory of measurement: central role of instruments.
 - Ideally, instruments *convert/transduce* an input phenomenon into a phenomenon that an observer should be able to access without uncertainty.



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■ Measurement system (MS)

- (physical) support m [the scale]
 - empirical structure $\mathcal{E} = \langle U, R_1, \dots, R_n \rangle$ [$\{s_0, s_1, s_2, s_3\}; s_0 < s_1 < s_2 < s_3$]
 - interaction function: $\kappa : D \rightarrow U$ [D : entities that can interact with m]
 - symbolic struct. $\mathcal{S} = \langle V, S_1, \dots, S_n \rangle$ [$\{0\text{kg}, 1\text{kg}, 2\text{kg}, 3\text{kg}\}; 0\text{kg} < 1\text{kg} < 2\text{kg} < 3\text{kg}$]
 - symbolization function: $\lambda : U \rightarrow V$
-
- $\lambda(\kappa(o)) = 1\text{kg}$
 - U gives the *resolution* of the MS: $o \approx o'$ iff $\kappa(o) = \kappa(o')$
 - R_i s induce relations on entities: $R_i^D(o_1, \dots, o_n)$ iff $R_i(\kappa(o_1), \dots, \kappa(o_n))$

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- Metrology: *inter-subjectivity* is obtained via
 - *measurement standard* (mST): set of reference objects with conventionally assigned labels and relations;
 - *measurement procedures*: how instruments should be used and in which environmental conditions;
 - *calibration*: an instrument is calibrated wrt an mST if the measurement of reference objects matches the conventions;
 - *alignment*: calibration of instruments wrt the same mST.

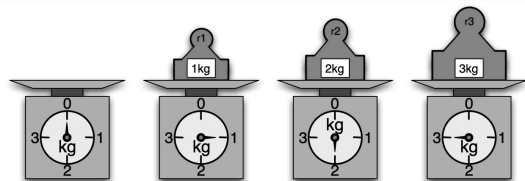
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■ *Measurement standard* (mST)

- set R of *reference entities*: $\{r_0, r_1, r_2, r_3\}$; [problem with 'null object' r_0]
- *symbolic str.* $\mathcal{R} = \langle M, S_1^M, \dots, S_n^M \rangle$; $[\{0\text{kg}, 1\text{kg}, 2\text{kg}, 3\text{kg}\}; 0\text{kg} < 1\text{kg} < 2\text{kg} < 3\text{kg}]$
- $\alpha : R \rightarrow M$ is a one-to-one function that conventionally assigns to each object in R a symbol in M : $\alpha(r_n) = n\text{kg}$

■ *Calibration*: MS $\langle m, \mathcal{E}, \kappa, \mathcal{S}, \lambda \rangle$ is calibrated wrt mST $\langle R, \mathcal{R}, \alpha \rangle$ iff:

- $\mathcal{S} = \mathcal{R}$ (or one-to-one relation between $\mathcal{S}$ and $\mathcal{R}$);
- for each $r \in R$, $\lambda(\kappa(r)) = \alpha(r)$ and
- for each $r_1, \dots, r_n \in R$, $S_i(\lambda(\kappa(r_1)), \dots, \lambda(\kappa(r_n)))$ iff $S_i^M(\alpha(r_1), \dots, \alpha(r_n))$



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- Values (in quantitative cases).
 - Require a *metrological* system (measurement standards connected through traceability chains) and calibration processes.
 - Relational form of information: the ratio of the measured property to the chosen unit.
- Quantitative *vs.* Qualitative analysis.
 - Measuring properties *vs.* quantities.
 - Continuous *vs.* discrete classifications (+ discretization).
 - Structure of properties/quantities (nominal scales without structure).

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- Philosophical debate: data as *given* vs. data as *made*.
 - Instruments discern *real* properties:
 - an object weights **1kg** because it is **1kg** heavy.
 - Instruments offer a *perspective* on reality:
 - an object is **1kg** heavy because it weights **1kg**.
 - an object is classified as 'being 1kg heavy' if by measuring it with a (calibrated) scale the label of the resulting internal state of the scale is '**1kg**';
 - Beliefs vs. Knowledge (Epistemic Logics).

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- A similar debate exists in cognitive sciences (Matthen 2005).
 - Are distal stimuli classed together because they share some physical property?
 - An object looks red because it is red.
 - Or, on the basis of some environmentally useful response that they evoke in sensory systems?
 - An object is red because it looks red.

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- Empirical/epistemological dimension of data.
 - Myth of **raw data**: measurements, observations, and analyses have a subjective nature that transcends the conceptual apparatus necessary to represent the domain (operationalist/constructivist stance about data).
 - *Direct vs. indirect* data: measurement is not only an empirical process, it can also include a *computational stage* that functionally calculates the result based on the measurements of other quantities or qualities.
 - Physical **transduction** *vs.* **computation** or calculation.
 - To share measures, to allow measures to “convey the same information to different observers”, it is not necessary to assume an objective reality.
 - It is enough to calibrate the instruments relying on mSTs conventionally established with the intended use of providing some (reliable) predictions (instruments are intentionally tuned with the ‘out there’).

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- How to represent provenance.
 - Inference/entailment.
 - Sufficient conditions (and assumed TBox).
 - Direct vs. indirect data (data resulting from “computations”).
 - Difficulty to represent equations and numerical rules in a logic.
 - Meta-level annotations(-graphs).
 - Prevents a uniform approach where the information about the provenance of data is intrinsic to the schema rather than an external annotation.
 - Additional arguments.
 - 60KG(*john*) *vs.* 60KG(*john*, *scale#1*, *t*)

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■ Reification (W3C Pattern for representing n -ary relations).

- $R(a, b, c)$ vs. $R^*(r) \wedge ARG_1(a, r) \wedge ARG_2(b, r) \wedge ARG_3(c, r)$
- $60KG(john, scale\#1, t)$ vs.
 $60KG^*(w) \wedge ARG_1(john, w) \wedge ARG_2(scale\#1, w) \wedge ARG_3(t, w)$
 $60KG^*(w) \wedge SUBJ(john, w) \wedge INSTR(scale\#1, w) \wedge TIME(t, w)$ [thematic roles]
 $WEIGHT(w) \wedge VALUE(60kg, w) \wedge SUBJ(john, w) \wedge INSTR(scale\#1, w) \wedge TIME(t, w)$
 $WEIGHT(w) \wedge VALUE(60kg, w) \wedge SUBJ(w_{john}, w) \wedge INSTR(scale\#1, w) \wedge TIME(t, w)$
- New concepts and relations that must be plugged into the ontology.
- Increased number of instances and relationships (for the different technical RDF-oriented implementations, see (Massari et al. 2025)) but
- relations between data and/or measurement processes may be added (including justification or grounding relations);
- ? Suppose $SCARLET(x) \rightarrow RED(x)$ and $SCARLET(a)$. [granularity/resolution]
Is it necessary to reify both $SCARLET(a)$ and $RED(a)$?

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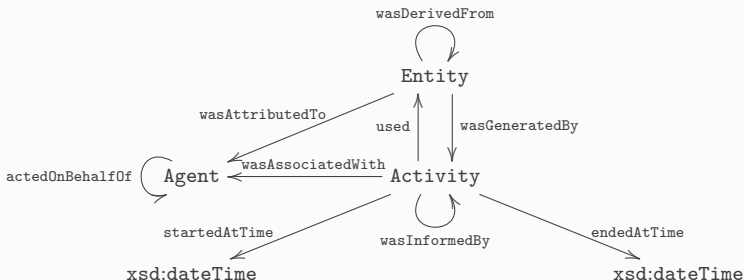
- Semantic Web oriented works about the nature/provenance of data.
 - QUDT (Quantity, QuantityValue, QuantityKind; Unit; SystemOfUnits).
(<https://www.qudt.org/>)
 - SSN: Semantic Sensor Network ontology (W3C Reccom. + OGC Standard).
(www.w3.org/TR/vocab-ssn/)
 - **PROV ontology.**
(www.w3.org/TR/prov-o/)
 - **SOSA: Sensor, Observation, Sample, and Actuator ontology.**
(Janowicz et al. 2018)
 - Core ontology at the basis of Semantic Sensor Network (SSN) ontology.
 - STAD: An Ontology Design Pattern and Ontology for the Semantic Representation of Aggregate Spatial and Temporal Data.
(Wiafe-Kwakye et al. 2022)
 - Descriptions and Situations design pattern.
(Gangemi & Mika 2003)

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- More foundational works.
 - Ontologies of observations.
(Kuhn 2009; Masolo 2010, Masolo et al. 2018, Probst 2008)
 - Information Artifact ontology.
(Smith & Ceusters 2015)
 - Ontology for Biomedical Investigations.
(Brinkman et al. 2010)
 - DataTop ontology.
(Batrancourt et al. 2010)
 - Scientific experiments and measurement.
(Barcellos et al. 2014, da Cruz et al. 2012).

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- PROV Ontology: Starting Point classes and properties.



Provenance chains:

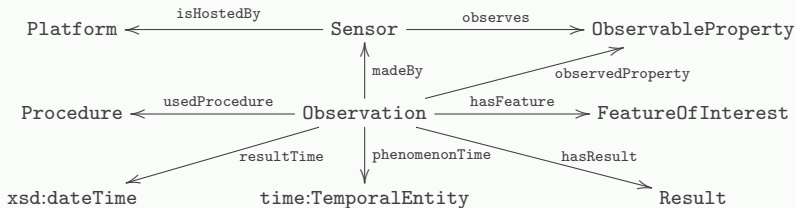
- Activity–Entity via `used` and `wasGeneratedFrom`;
- Activity–Activity via `wasInformedBy`;
- Entity–Entity via `wasDerivedFrom`.

■ PROV Ontology

- **Entity**: a physical, digital, conceptual, or other kind of thing with some fixed aspects; entities may be real or imaginary.
- **Activity**: something that occurs over a period of time and acts upon or with entities; it may include consuming, processing, transforming, modifying, relocating, using, or generating entities.
- **Agent**: something that bears some form of responsibility for an activity taking place, for the existence of an entity, or for another agent's activity.
- **wasInformedBy** suggests that the informed Activity used an Entity that was generated by the informing Activity, but the Entity itself is unknown or is not of interest.
- **wasDerivedFrom**: a derivation is a transformation of one entity into another.

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- SOSA Observation perspective (simplified) [+ actuation & sampling]



- Mappings with the PROV ontology:

- **Observation** $\sqsubseteq$ **prov:Activity**
- **Sensor** $\sqsubseteq$ **prov:Agent**
- **Procedure** $\sqsubseteq$ **prov:Plan** (prov:Entity that represents a set of actions or steps intended by one or more agents to achieve some goals.)
- **FeatureOfInterest** $\sqsubseteq$ **prov:Entity**

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■ **FeatureOfInterest**

- The thing whose property is being estimated or calculated in the course of an **Observation** to arrive at a **Result** (*e.g.*, a tree).

■ **ObservableProperty**

- An observable quality (property, characteristic) of a **FeatureOfInterest**. (*e.g.*, the height of a tree).
 - Subclass of **ssn:Property**. A quality of an entity. An aspect of an entity that is intrinsic to and cannot exist without the entity.
- One observable property can apply to acts of observation concerning different features of interest, at different times, or using different procedures or sensors.

■ **Observation**

- Act of carrying out an (observation) **Procedure** to estimate or calculate a value of an **ObservableProperty** of a **FeatureOfInterest**. An observation involves a **Sensor** and yields a **Result**.

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■ Sensor

- Device, agent (including humans), or software (simulation) involved in, or implementing, a **Procedure**. Sensors respond to a stimulus, e.g., a change in the environment, or input data composed from the **Results** of prior **Observations**, and generate a **Result**. **Sensors** can be hosted by **Platforms**. They can be thought of as implementations (of parts) of an observation procedure.

■ Procedure

- A workflow, protocol, plan, algorithm, or computational method specifying how to make an **Observation** (...). A **Procedure** is re-usable, and might be involved in many **Observations** (...). It explains the steps to be carried out to arrive at reproducible **Results**.

■ **phenomenonTime** vs. **resultTime**

- time that the result of an act of observation applies to the feature of interest vs. instant of time when the act was completed.
- **resultTime** is the instant of time when the **Observation** activity was completed. It is a datatype property, while **phenomenonTime** (interval or instant) is an object property based on OWL-Time ontology.

■ Nature of the measurands.

- SOSA: FeatureOfInterest (entity) *vs.* ObservableProperty (individual prop.).
- International Vocabulary of Metrology
(3rd ed. (VIM3), 2006-08-01 - 4th ed. (VIM4) still draft)
 - (**quantity** <general>) (e.g., length, electric charge)
(VIM3) property of a phenomenon, body, or substance, to which a number can be assigned with respect to a reference
(VIM4) property whose instances can be compared by ratio or only by order
 - (**individual quantity**) (e.g., electric charge of the proton, e)
(VIM4) instance of a general quantity
 - (**measurand**) (VIM 3/4) quantity intended to be measured
- Representational theory of measurement: state of an entity.
- Empirical theory of measurement: an entity / the quality of an entity.
 - “An empirical property of an object is associated with a mode of empirical interaction of the object with its environment.” (Mari et al. 2023, p.111)
- Are both (qualities of) objects and (qualities of) events measurands?

- Nature of the measurements/data/observations.
 - True (entertained) propositions corresponding to facts, states of affairs, tropes or individual properties.
 - Relies on objective (independent from cognition) and the stable (independent from time) properties.
 - Not necessarily veridical propositions, private sensations or qualia that depend on single observers/sensors can be accepted.
 - Operationalism or conventionalism: properties as created by the application of measurement/perceptive processes or conventions.
 - Linguistic entities like expressions, symbols, etc.

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- Nature of the measurements/data/observations.
 - SOSA: Result (non clear ontological nature)
(value 20 as the height of a certain tree together with the unit, *e.g.*, Meter)
 - VIM:
 - **(quantity value; value of a quantity; value)**
(VIM3) number and reference together expressing magnitude of a quantity
(VIM4) individual quantity identified as the product of a number and a measurement unit or on an ordinal scale
 - **(measurement result; result of measurement)** set of quantity values being attributed to a measurand together with any other available relevant information
 - **(measured quantity; value measured value of a quantity; measured value)**
quantity value representing a measurement result

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- Nature of the measurements/data/observations.
 - Mathematical tuples $\langle\langle a, b, c \rangle, R\rangle$.
 - Time: are data in time? When they exist (existential conditions)? Can they exist at different times (persistence criteria)?
- Nature of the sensor/instrument.
 - SOSA include humans among the sensors.

- *Reading Comprehension Ability (RCA).* (Mari et al. 2023)
 - A reader reads a textual passage, and is then asked one or more questions (called items) about how well they understand the contents of the passage.
 - The reader responds to each item by writing a response or selecting an option from a predetermined set of options. Thus, the reader's RCA is transduced to the responses to the items.
 - The correctness of each item response is (automatically) rated.
 - Indication of reader's RCA is given by the pattern of correct/incorrect responses for the set of items, summarized in terms of the number (or percentage) of test items that the reader answered correctly (sum score).
 - Sum score is an indication at the macro level of the reader's comprehension, whereas each individual item response is an indication at the microlevel of the reader's comprehension of the question asked in the item.
 - This indication is limited in its interpretation to the specific set of items administered. Various methods are available to allow interpretation beyond specific tests to generate a value on an instrument-independent RCA scale.

- *Reading Comprehension Ability* (RCA). (Mari et al. 2023)
 - Norm-referenced approach: it relates the RCA values to the sum score distribution for a chosen reference population of readers:
 - representative sample of individual readers from a specified population;
 - take a test and generates a sample of sum scores;
 - statistical analysis on sum scores: mean and standard deviation.
 - Criterion-referenced approach: RCA scale values are related to specific reference reading comprehension criteria.
- Is it possible to relate the account of an RCA test to the traditional idea of a physical instrument such as a scale or thermometer?

- Validity plays an important role in evaluating tests.
 - a test measures what it claims to measure
 - present state of available evidence, as judged by a particular community
- “validity is an integrated evaluative judgment of the degree to which empirical evidence and theoretical rationales support the adequacy and appropriateness of inferences and actions based on test scores” (Messick, 1989: p. 13) cited in (Mari et al. 2023).

■ Qualitative Social Research

- “Qualitative research is a situated activity that locates the observer in the world. It consists of a set of interpretative, material practices that make the world visible. These practices transform the world. They turn the world into a series of representations, including fieldnotes, interviews, conversations, photographs, recordings, and memos of the self. At this level, qualitative research involves an interpretive, naturalistic approach to the world. This means that qualitative researchers study things in their natural setting, attempting to make sense of, or interpret, phenomena in terms of the meanings people bring to them”
(Denzin & Lincoln 2017)
- Naturally occurring phenomena: analysing non replicable events (vs. experiments) yet in depth and detail – ‘the world as it happens’
- Methods: e.g. ethnography, in-depth interviews, Conversation Analysis (CA), focus groups (possibly combined).
- No measurement standards but examples accepted by given communities of experts. ~Theory of Exemplars where (indirect) measuring is based on socially recognized evaluation of measurement and a distance/similarity relation.

- *Relational* notion of data. (Leonelli 2015)
 - Data is a relational category and “any object can be considered as a datum as long as (1) it is treated as potential evidence for one or more claims about phenomena and (2) it is possible to circulate it among individuals.”
 - Data “as product of research activities, ranging from artifacts such as photographs to symbols such as letters or numbers, which is collected, stored, and disseminated in order to be used as evidence for knowledge claims.”
“Data can include experimental results, field observations, samples of organic materials, results of simulations and mathematical modeling, and even specimens.”
 - “This does not mean that whoever gathers data already knows how they might be used. Rather, what matters is that observations or measurements are collected with the expectation that they may be used as evidence for claims about the world in the future.”
 - “What counts as data depends on who uses them, how, and for which purposes.”

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■ Degree of classification

- Fuzzy logics: degree of truth.
- Prototype Theory and Theory of Exemplars (cognitive science): degree of categorization to model typicality effects.

■ Error approach.

- “Due to its experimental component, measurement is unavoidably affected by errors, understood as the difference between the measured value and the true value. Under the assumption of the unknowability of true values (...) this sharp characterization has been sometimes weakened by instead considering “conventional true values””
(Mari et al. 2023)

■ Uncertainty approach.

- Focus “on the state of partial knowledge of the measurer, who designs and performs the measurement for the explicit purpose of gaining information on the measurand, with the acknowledgment that “complete” information (whatever this may actually mean) is unobtainable even by the best possible measurement.”
(Mari et al. 2023)

■ Complications

- “[P]ressure gauges do not tell you which (if any) tire it is that they represent the pressure of. If you want to know which tire (if any) the gauge represents (...) you have to look, not at the gauge, but at the external connections between gauge and world (...) Gauges do not supply (...) information about these external connections (...) this is why one cannot know, at least not by introspection, what object (or whether there is an object) one is experiencing”
(Dretske 1995)
- What if I have access only to the internal states of the instrument without seeing what I’m measuring? (E.g., in perceptive processes)

■ Complications

- How to check that different instruments measure (point to) the same entity (data aggregation)?
- How to check which entity the instrument is measuring (*e.g.*, which tire the gauge is connected to)?
- How to individuate and re-identify entities (*e.g.*, tires)?

Vicious circles?

- The stability of the measurement standards and instruments cannot be assumed.
- To empirically justify the stability of reference framework one needs to diachronically compare instruments and reference objects; are additional instruments needed?
- Similar questions in cognitive science: (Pylyshyn 2007) object files and FINST.
 - Initial individuation and tracking of objects is not conceptual, it is not based on the classification under concepts, but on a lower level mechanism built into the visual system called FINST.

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- Moving from outside to inside the instrument we don't know what exactly objects are.
- Objects are central for coordination/interaction, then they could be understood looking at their role in the solution of the coordination problem.
- Can we do something similar to data?

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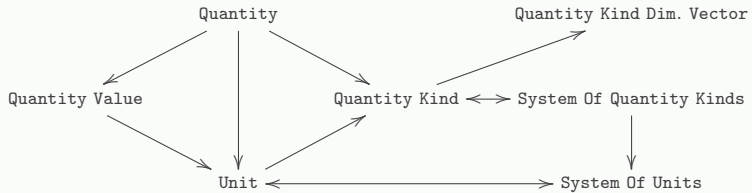
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ADDITIONAL SLIDES

■ QUDT



■ The Information Artifact Ontology (IAO)

- IAO is based on the copying and aboutness relations: an information content entity (ICE) is, roughly, an entity that can be copied (in the way, for example, that a gene sequence or a computer file can be copied) and that is about something. The aboutness relation itself has its roots in the mental realm – an information content entity e is about something, again roughly speaking, in virtue of the fact that the mental acts which gave rise to or which are used to interpret or communicate with e are about something.
- An ICE is thus conceived as an entity which is about something in reality and which can migrate or be transmitted (for example through copying) from one entity to another.
- INFORMATION CONTENT ENTITY =def. an ENTITY which is (1) GENER-ICALLY DEPENDENT on (2) some MATERIAL ENTITY and which (3) stands in a relation of ABOUTNESS to some ENTITY.

■ The Information Artifact Ontology (IAO)

- INFORMATION ARTIFACT =def. an ARTIFACT whose function is to bear an INFORMATION QUALITY ENTITY. (Examples: a hard drive, a traffic sign, a printed form, a passport, a currency note, an RFID chip, a SIM card).
- All information artifacts are material entities. While every ICE is dependent upon some material entity that is its bearer ICEs themselves are not material entities.
- We shall presuppose in what follows that information artifacts do not bear information in and of themselves, but only because cognitive subjects associate representations of certain sorts with the patterns which they manifest. We thus view the aboutness that is manifested by information content entities in accordance with the doctrine of the 'primacy of the intentional' (Chisholm, 1984), according to which the aboutness of those of our representations formulated in speech or writing (or in their printed or digital counterparts) is to be understood by reference to the cognitive acts with which they are or can in principle be associated.